Muhammad Asad Zaighum

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Summary

My research lies at the intersection of harmonic analysis and functional analysis, with a particular focus on analysis associated with non-standard function spaces. I bring nearly a decade of university-level teaching experience, developed alongside an active research career following the completion of my Ph.D. in Mathematics.

My work has been published in reputable international journals, and I have supervised graduate students, including Ph.D. candidates. Beyond teaching and research, I have served as a department chair, where I gained substantial experience in academic leadership, program development, and institutional management.

I am committed to advancing high-quality research, supporting academic excellence and contributing meaningfully to the strategic growth of academic institutions.

Academic Qualifications

- **Abdus Salam School of Mathematical Sciences, Government College University Lahore** *2009 - 2014*
  Ph.D. in Mathematics Lahore, Pakistan
  - Dissertation: Kernel Operators in Some New Function Spaces
  - Advisor: Professor Alexander Meskhi
- **COMSATS Institute of Information and Technology** *2005 - 2009*
  BS Mathematics Islamabad, Pakistan
  - Project: Perturbation Methods in Physiological Flows
  - Advisor: Professor Saleem Asghar

Professional Experience

- **Heriot-Watt International Faculty, K. Zhubanov Aktobe Regional University [🌐]** *December 2025 - Present*
  Associate Professor Aktobe, Kazakhstan
  - Teaching mathematics at Global College, Heriot-Watt University, foundation-level Mathematics course (SCQF Level 7).
- **Nazarbayev University [🌐]** *August 2024 - December 2025*
  Postdoctoral Research Assistant Astana, Kazakhstan
  - Conducting research under the project titled "Qualitative analysis for nonlocal and fractional models".
- **Riphah International University[🌐]** *September 2020 - On Leave Since August 2024*
  Head of Department, Mathematics and Statistics Islamabad, Pakistan
  - Overseeing departmental operations and curriculum development.
  - Managing academic programs and faculty coordination.
- **Riphah International University[🌐]** *September 2018 - On Leave Since August 2024*
  Associate Professor, Mathematics and Statistics Islamabad, Pakistan
  - Teaching courses for undergraduate and postgraduate students.
  - Supervising Ph.D. and M.Phil. students in mathematics.
- **Pontificia Universidad Javeriana[🌐]** *May 2015 - February 2016*
  Postdoctoral Investigator Bogota, Colombia
  - Conducted research with Professor Humberto Rafeiro under the project titled "Study of boundedness of operators in generalized Morrey spaces ".
  - Presented talks at international conferences during the period.
- **Riphah International University[🌐]** *April 2014 - August 2018*
  Assistant Professor, Mathematics and Statistics Islamabad, Pakistan
  - Developed and taught undergraduate and graduate mathematics courses.
  - Supervised student research projects and theses.

Honors and Awards

- **Research Performance Prize** *November 2013*
  Abdus Salam School of Mathematical Sciences, GCU Lahore
  - Awarded for outstanding research work during Ph.D.

- **First Prize, Young Speaker** *March 2013*
  6th World Conference on 21st Century Mathematics, Abdus Salam School of Mathematical Sciences
  - Recognized for exceptional presentation in the Young Speaker category.

- **Award for Commendable Performance** *November 2010*
  Abdus Salam School of Mathematical Sciences, GCU Lahore
  - Awarded for excellent performance in first-year Ph.D. coursework.

- **Chancellor's Gold Medal** *2009*
  COMSATS Institute of Information and Technology, Islamabad
  - Awarded for academic excellence in BS Mathematics.

- **Institute's Gold Medal** *2009*
  COMSATS Institute of Information and Technology, Islamabad
  - Recognized for outstanding academic performance in BS Mathematics.

Research Supervision

- **Ph.D. Supervision**
  Completed: 6, Ongoing: 1
  - Supervised 6 Ph.D. students to completion in topics including Geometric Function Theory and Harmonic Analysis.
  - Currently supervising one Ph.D. student on the topic of Herz-Hardy type spaces with variable exponent.

- **M.Phil. Supervision**
  Completed: 35, Ongoing: 0
  - Supervised 35 M.Phil. students in areas such as advanced topics in function space theory.

Administrative Roles

- **In-Charge, Undergraduate Programs** *Spring 2016 - Fall 2020*
  Riphah International University
  - Managed undergraduate mathematics programs and curriculum development.

- **Timetable In-Charge** *Spring 2016 - 2020*
  Riphah International University
  - Coordinated scheduling for mathematics programs.

- **In-Charge, Student Affairs** *Fall 2014 - Spring 2015*
  Riphah International University
  - Oversaw student affairs for the Department of Basic Sciences.

Research Metrics

Database	**Scopus**	**Web of Science**
Indexed Articles	34	31
h-index	11	10

(Updated April 2026)

[1]

[J.1] Orazbayev, A., & Zaighum, M. A. (2025). **Generalized potential operators in weighted grand Herz spaces**. *Georgian Mathematical Journal*. (Impact Factor 0.7). [Scopus]

[J.2] Nursultanov, E., Suragan, D., & Zaighum, M. A. (2025). **On the boundedness of Riesz potential operators: insights from net spaces**. *Constructive Mathematical Analysis*. (Impact Factor 1.8). [Scopus]

[J.3] Tulendinova, A., Ghorbanalizadeh, A., & Zaighum, M. A. (2025). **On Grand Morrey Sequence Spaces**. *Journal of Pseudo-Differential Operators and Applications*. (Impact Factor 1.3). [Scopus]

[J.4] Kainat, A., Nafis, H., & Zaighum, M. A. (2025). **Boundedness of Parametric Marcinkiewicz Integral Operator on Grand Variable Exponent Herz Spaces**. *Journal of Mathematical Sciences*. [Scopus]

[A.1] Suragan, D., & Zaighum, M. A. (2025). **Grand Net Spaces and Applications to Integral Operators**. *Accepted: Rendiconti del Circolo Matematico di Palermo Series 2*.

[J.5] Shabbir, F., & Zaighum, M. A. (2025). **Boundedness of Marcinkiewicz integral operators on grand variable Herz–Hardy spaces**. *Georgian Mathematical Journal*. (Impact Factor 0.7). [Scopus]

[J.6] Shabbir, F., & Zaighum, M. A. (2024). **On the Boundedness of Sublinear Operators on Grand Herz–Hardy Spaces with Variable Exponent**. *Mediterranean Journal of Mathematics*, 21(2), 51. (Impact Factor 1.2). [Scopus]

[J.7] Nafis, H., Rafeiro, H., & Zaighum, M. A. (2022). **Boundedness of Multilinear Calderón-Zygmund Operators on Grand Variable Herz Spaces**. *Journal of Function Spaces*, Article ID 4845507, 11 pages. (Impact Factor 1.281). [Scopus]

[J.8] Nafis, H., Rafeiro, H., & Zaighum, M. A. (2021). **Boundedness of the Marcinkiewicz integral on Grand variable Herz spaces**. *Journal of Mathematical Inequalities*, 15(2), 739–753. (Impact Factor 1.219). [Scopus]

[J.9] Nafis, H., Rafeiro, H., & Zaighum, M. A. (2020). **A note on the boundedness of sublinear operators on grand variable Herz spaces**. *Journal of Inequalities and Applications*, 2020(1), 1. (Impact Factor 1.136). [Scopus]

[J.10] Mehmood, A., Al-Shomrani, M., Zaighum, M. A., & Abdullah. (2020). **Characterization of Soft S-Open Sets in Bi-Soft Topological Structure Concerning Crisp Points**. *Mathematics*, 8(12), 2100. (Impact Factor 2.258). [Scopus]

[J.11] Wei, J., Imran, M., Iqbal, M. A., & Zaighum, M. A. (2020). **Zagreb-Type Indices of Vertex Join and Edge Join of Graphs**. *Journal of Chemistry*, 2020, 1–10. (Impact Factor 2.506). [Scopus]

[J.12] Iqbal, M. A., Imran, M., Siddiqui, M. K., & Zaighum, M. A. (2020). **Computing Molecular Topological Descriptors of Polymeric Networks Modeled by Sierpinski Networks**. *Polycyclic Aromatic Compounds*, 1–20. (Impact Factor 3.774). [Scopus]

[J.13] Imran, M., Iqbal, M. A., Liu, Y., Baig, A. Q., Khalid, W., & Zaighum, M. A. (2020). **Computing eccentricity-based topological indices of 2-power interconnection networks**. *Journal of Chemistry*, 2020, 1–11. (Impact Factor 1.068). [Scopus]

[J.14] Iqbal, M. A., Imran, M., & Zaighum, M. A. (2020). **Eccentricity based topological indices of siloxane and POPAM dendrimers**. *Main Group Metal Chemistry*, 43(1), 92–98. (Impact Factor 2.506). [Scopus]

[J.15] Khan, S., Hussain, S., Zaighum, M. A., & Darus, M. (2019). **A subclass of uniformly convex functions and a corresponding subclass of starlike function with fixed coefficient associated with q-analogue of Ruscheweyh operator**. *Mathematica Slovaca*, 69(4), 825–832. (Impact Factor 0.490). [Scopus]

[J.16] Meskhi, A., Rafeiro, H., & Zaighum, M. A. (2019). **On the boundedness of Marcinkiewicz integrals on continual variable exponent Herz spaces**. *Georgian Mathematical Journal*, 26(1), 105–116. (Impact Factor 0.7). [Scopus]

[1] [S.] denotes preprint; [J.] denotes published journal article; [A.] accepted article.

[J.17] Meskhi, A., Rafeiro, H., & Zaighum, M. A. (2018). **Interpolation of an analytic family of operators on variable exponent Morrey spaces**. *Hiroshima Mathematical Journal*, 48(3), 335–346. (Impact Factor 0.250). [Scopus]

[J.18] Kokilashvili, V., Meskhi, A., & Zaighum, M. A. (2018). **Sharp weighted bounds for fractional integrals via the two-weight theory**. *Banach Journal of Mathematical Analysis*, 12(3), 673–692. (Impact Factor 0.927). [Scopus]

[J.19] Hussain, S., Khan, S., Zaighum, M. A., & Darus, M. (2018). **Applications of a q-Salagean type operator on multivalent functions**. *Journal of Inequalities and Applications*, 2018(1), 301. (Impact Factor 1.136). [Scopus]

[J.20] Ashiq, M., Imran, T., & Zaighum, M. A. (2017). **Defining relations of a group $\Gamma = G^{3,4}(2,\mathbb{Z})$ and its action on real quadratic field**. *Bulletin of the Iranian Mathematical Society*, 43(6), 1811–1820. (Impact Factor 0.313). [Scopus]

[J.21] Ashiq, M., Imran, T., & Zaighum, M. A. (2018). **Actions of $\Delta(3,n,k)$ on projective line**. *Transactions of A. Razmadze Mathematical Institute*, 172(1), 1–6. (ESCI-Indexed).

[J.22] Meskhi, A., Rafeiro, H., & Zaighum, M. A. (2018). **Central Calderón–Zygmund operators on Herz-type Hardy spaces of variable smoothness and integrability**. *Annals of Functional Analysis*, 9(3), 310–321. (Impact Factor 0.30). [Scopus]

[J.23] Kokilashvili, V., Meskhi, A., & Zaighum, M. A. (2017). **Sharp weighted bounds for one-sided operators**. *Georgian Mathematical Journal*, 24(2), 227–240. (Impact Factor 0.7). [Scopus]

[J.24] Hussain, S., Rasheed, A., Zaighum, M. A., & Darus, M. (2017). **A Subclass of Analytic Functions Related to Uniformly Convex and Starlike Functions**. *Journal of Function Spaces*, 2017, Article ID 9010964. (Impact Factor 0.426). [Scopus]

[J.25] Meskhi, A., Rafeiro, H., & Zaighum, M. A. (2017). **Complex interpolation on variable exponent Campanato spaces of order k**. *Complex Variables and Elliptic Equations*, 62(6), 795–813. (Impact Factor 0.806). [Scopus]

[J.26] Meskhi, A., & Zaighum, M. A. (2016). **Weighted Kernel Operators in $L^{p(x)}(\mathbb{R}^+)$ spaces**. *Journal of Mathematical Inequalities*, 10(3), 623–639. (Impact Factor 1.158). [Scopus]

[J.27] Kokilashvili, V., Meskhi, A., & Zaighum, M. A. (2016). **Sharp weighted bounds for multiple operators**. *Transactions of A. Razmadze Mathematical Institute*, 170(1), 75–90. (ESCI-Indexed).

[J.28] Meskhi, A., Rafeiro, H., & Zaighum, M. A. (2016). **Interpolation on variable Morrey spaces defined on quasi-metric measure spaces**. *Journal of Functional Analysis*, 270(10), 3946–3961. (Impact Factor 1.637). [Scopus]

[J.29] Meskhi, A., & Zaighum, M. A. (2014). **On the Boundedness of maximal and potential operators in variable exponent amalgam spaces**. *Journal of Mathematical Inequalities*, 8(1), 123–152. (Impact Factor 1.158). [Scopus]

[J.30] Kokilashvili, V., Meskhi, A., & Zaighum, M. A. (2013). **Positive kernel operators in $L^{p(x)}$ spaces**. *Positivity*, 17(4), 1123–1140. (Impact Factor 0.833). [Scopus]

[J.31] Kokilashvili, V., Meskhi, A., & Zaighum, M. A. (2013). **Weighted kernel operators in variable exponent amalgam spaces**. *Journal of Inequalities and Applications*, 2013, 173. (Impact Factor 1.136). [Scopus]

[J.32] Meskhi, A., & Zaighum, M. A. (2012). **On the boundedness of product kernel operators with measures**. *Georgian Mathematical Journal*, 19(3), 533–557. (Impact Factor 0.551). [Scopus]

Skills

- **Mathematical & Statistical Tools:** LaTeX, Maple, Microsoft Office
- **Research Skills:** Harmonic Analysis, Theory of Function Spaces, Grand Function Spaces, Weighted Norm Inequalities, Complex Interpolation

Certifications

- **GRE International Mathematics** *63 Percentile*

Conferences and Workshops

- **15th ISAAC Congress** *July 2025*
  Nazarbayev University, Astana, Kazakhstan
  - Delivered a talk titled "Hardy-Littlewood Sobolev theorem on net spaces" on July 24, 2025.
- **Traditional Int'l April Mathematical Conference in Honor of the Kazakhstan Day of Science** *April 2025*
  Institute of Mathematics and Mathematical Modelling, Almaty, Kazakhstan
  - Delivered a Plenary talk titled "Grand Net Spaces and Application" on April 2, 2025.
  - Delivered a Session talk titled "On the Boundedness of Multilinear Calderon-Zygmund Operators" on April 1, 2025.
- **Conference on Actual Problems of Analysis, Differential Equations and Algebra -2025** *January 2025*
  Eurasian National University, Astana, Kazakhstan
  - Delivered a talk titled "On a Generalization of Net Spaces" on January 7, 2025.
- **ICOMET 2023** *March 2023*
  IBA Sukkur
  - Keynote Speaker at the conference held on March 17–18, 2023.
- **Winter Workshop on Mathematical Sciences** *December 2017*
  Abdus Salam School of Mathematical Sciences (ASSMS), Lahore, Pakistan
  - Invited Speaker at the workshop.
- **Coloquio de Mathematicas** *September 2015*
  Pontificia Universidad Javeriana, Bogota, Colombia
  - Delivered a talk titled "Sharp Weighted Bounds for One-Sided Operators" on September 15, 2015.
- **XX Congresso Colombiano de Mathematicas** *July 2015*
  Universidad Nacional de Colombia, Sede Manizales, Colombia
  - Delivered a talk titled "Sharp Weighted Bounds for One-Sided Operators" on July 22, 2015.
- **6th World Conference on 21st Century Mathematics** *March 2013*
  Abdus Salam School of Mathematical Sciences (ASSMS), Lahore, Pakistan
  - Delivered a talk titled "Product Kernel Operators with Measures" on March 6, 2013.
- **Scientific Seminars Series in Analysis** *October 2013*
  Abdus Salam School of Mathematical Sciences (ASSMS), Lahore, Pakistan
  - Delivered a talk titled "Boundedness of Integral Operators on Amalgams" on October 24, 2013.
- **Abdus Salam School of Mathematical Sciences (ASSMS)** *April 2013*
  Lahore, Pakistan
  - Delivered a talk titled "Kernel Operators in Lp(x) Spaces with Decay Condition at Infinity" in April 2013.
- **Abdus Salam School of Mathematical Sciences (ASSMS)** *October 2012*
  Lahore, Pakistan
  - Delivered a talk titled "Kernel Operators in Amalgam Spaces" in October 2012.
- **Scientific Seminars Series in Analysis** *November 2011*
  Abdus Salam School of Mathematical Sciences (ASSMS), Lahore, Pakistan
  - Delivered a talk titled "On the Boundedness of Multidimensional Product Kernel Operators in Lebesgue Spaces with Weights" on November 24, 2011.

Additional Information

Languages: English (Fluent), Urdu (Native)
Interests: Playing Cricket, Reading Tech Stuff

References

1. **Dr. Alexander Meskhi**
   Head of the Department of Mathematical Analysis, TSU A. Razmadze Mathematical Institute
   President of Georgian Mathematical Union
   Professor at Kutaisi International University
   Email: alexander.meskhi@kiu.edu.ge
   Phone: +995-593573895
   Relationship: Ph.D. Advisor

2. **Dr. Humberto Rafeiro**
   Associate Professor, Department of Mathematical Sciences
   United Arab Emirates University, Al Ain, UAE
   Email: hrafeiro@uaeu.ac.ae
   Relationship: Research Collaborator

3. **Dr. Durvudkhan Suragan**
   Professor, Department of Mathematics
   Nazarbayev University, Astana, Kazakhstan
   Email: durvudkhan.suragan@nu.edu.kz
   Relationship: Research Collaborator

List of Supervised Students by Dr. Muhammad Asad Zaighum

Ph.D. Students Supervised

S.No	Name	SAP/CMS	Date	Thesis Title
1	Hammad Nafis	F16C09P05001	25-08-2022	Boundedness of Some Operators on variable exponent function spaces
2	M Azhar Iqbal	F14C09P05001	17-02-2021	On Molecular Topological Properties of Chemical Graphs and Networks
3	Arif Mehmood	S15C09P05008	26-01-2021	Some Contribution to Bi soft Topological Spaces
4	Shahid Khan	S15C09P05001	17-06-2020	Linear Operators and Related topics in geometric Function Theory
5	Akhter Rasheed	S14C09P05003	24-09-2019	Application of Convolution and Subordinate Techniques in Geometric Function Theory
6	Tahir Imran	S14C09P05005	30-08-2018	Parameterization and Graphical Representation of Certain Groups